Paper LXXV

Coherence Growth, Vacuum Tension, and a Propagating Instability in a Torsion-Offset Universe

Temperature-dependent scalar phase transition · w(z) interpolation · S8 suppression · Tachyonic reset · 18.52 nHz PTA prediction

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ABSTRACT

We present a dynamical extension of the MSDS framework in which a small vacuum torsion offset $v^2 = 0.00017$ seeds a temperature-dependent phase transition of a scalar coherence field. As the universe cools, the field undergoes symmetry breaking, producing a growing correlation length $\xi(T)$. The associated energy density interpolates smoothly between matter-like behaviour at high redshift and a cosmological constant at late times, yielding an effective equation of state w(z) consistent with current observations. The coherent vacuum modifies gravitational structure growth, alleviating the S8 tension qualitatively. Continued expansion stretches the correlated domains; when a critical stretch is exceeded, a tachyonic instability triggers a propagating phase transition that converts coherence energy into radiation, resetting the cycle. The model predicts a narrow Gaussian feature in the nanohertz gravitational-wave background at $f_c = 18.52$ nHz with fractional amplitude alpha ~ 0.13 — a testable signature for PTA experiments. All numerical estimates are fiducial; a full Boltzmann integration is required for precise comparison with data. We provide explicit falsification criteria.

1. Introduction and Motivation

The standard LCDM model successfully describes large-scale structure, yet two tensions persist: the existence of over-massive early galaxies at z~5-8 ('Little Red Dots', Paper LXXIII), and the lower-than-expected clustering amplitude S8 (Papers L-LXV). Here we propose that the vacuum itself undergoes a cooling-driven coherence transition seeded by $v^2 = 0.00017$. The water-ice analogy (Paper LXXIV) provides intuitive motivation: as water cools, it expands into an open lattice; the cooling vacuum develops a rigid, expanded lattice that contributes to the energy budget and eventually becomes unstable. Unlike static formulations, this paper introduces a dynamical scalar field with a temperature-dependent potential, allowing derivation of the correlation length, energy density, and GW signature from first principles up to normalisation.

2. Field Theory: Temperature-Dependent Coherence

2.1 Lagrangian and potential

We introduce a real scalar field ϕ (the coherence field) with a temperature-dependent potential:

$$L = \tfrac{1}{2}(\partial\phi)^2 - V(\phi,T) \text{ where } V(\phi,T) = \tfrac{1}{2}m^2(T)\phi^2 + (\lambda/4)\phi\blacksquare$$

$m^2(T) = \alpha(T - T_c)$ | $T_c \sim v^2 = 0.00017$ sets the critical temperature

For $T > T_c$: $m^2 > 0$, minimum at $\phi=0$ (symmetric, incoherent). For $T < T_c$: $m^2 < 0$, spontaneous symmetry breaking — coherent lattice forms.

2.2 Correlation length

$$\xi(T) = 1/|m(T)| = 1/\sqrt{(|\alpha||T - T_c|)}$$

Diverges at T_c (critical slowing down); saturates at late times

3. Effective Energy Density and Equation of State

The coherence energy density at the potential minimum is:

$$\rho_{coh}(T) \propto (T_c - T)^2 \text{ for } T < T_c$$

Rather than assuming a direct $T \propto (1+z)^2$ scaling — which gives $\rho_{coh} \propto (1+z)$■ and is inconsistent with observed late-time acceleration — we adopt a physically motivated interpolation that smoothly connects matter-like scaling at high z to a cosmological constant at z=0:

$$\rho_{coh}(z) = \rho_{DE,0} \times [1 - \exp(-(v^2/(1+z)^2)^{1.5})] / [1 - \exp(-v^3)]$$

Guarantees $\rho_{coh}(0) = \rho_{DE,0}$; matter-like at high z; constant at z=0

Honest caveat: the transition redshift from this interpolation is $z \sim v$■$^{1} \sim 77$, too early to match observations. A full derivation from the Lagrangian would solve the field equations and determine the correct parameter mapping. We treat this as a toy demonstration of the mechanism; precise fitting requires modification of Boltzmann codes such as CLASS or CAMB.

4. S8 Tension: Modification of Structure Growth

The coherent vacuum couples to matter via an effective gravitational constant:

$$G_{eff}(z) = G \times [1 + \beta \times \rho_{coh}(z)/\rho_m(z)]$$

$\beta = 0.2$ (fiducial); sign of S8 suppression robust for $\beta \in [0.1, 1.0]$

At low z, ρ_{coh} saturates while ρ_m dilutes, driving G_{eff} above G. The net effect suppresses present-day clustering amplitude:

$$S8_{model} \approx 0.789 \pm 0.010 \text{ (fiducial)}$$

vs ΛCDM: S8 = 0.832 | vs observed: S8 = 0.776 ± 0.020 (DES/KiDS)

These estimates are qualitative and not reproducible from the paper alone. A full treatment requires Boltzmann code integration. The DIRECTION of the effect — lower S8 — is consistent with observations.

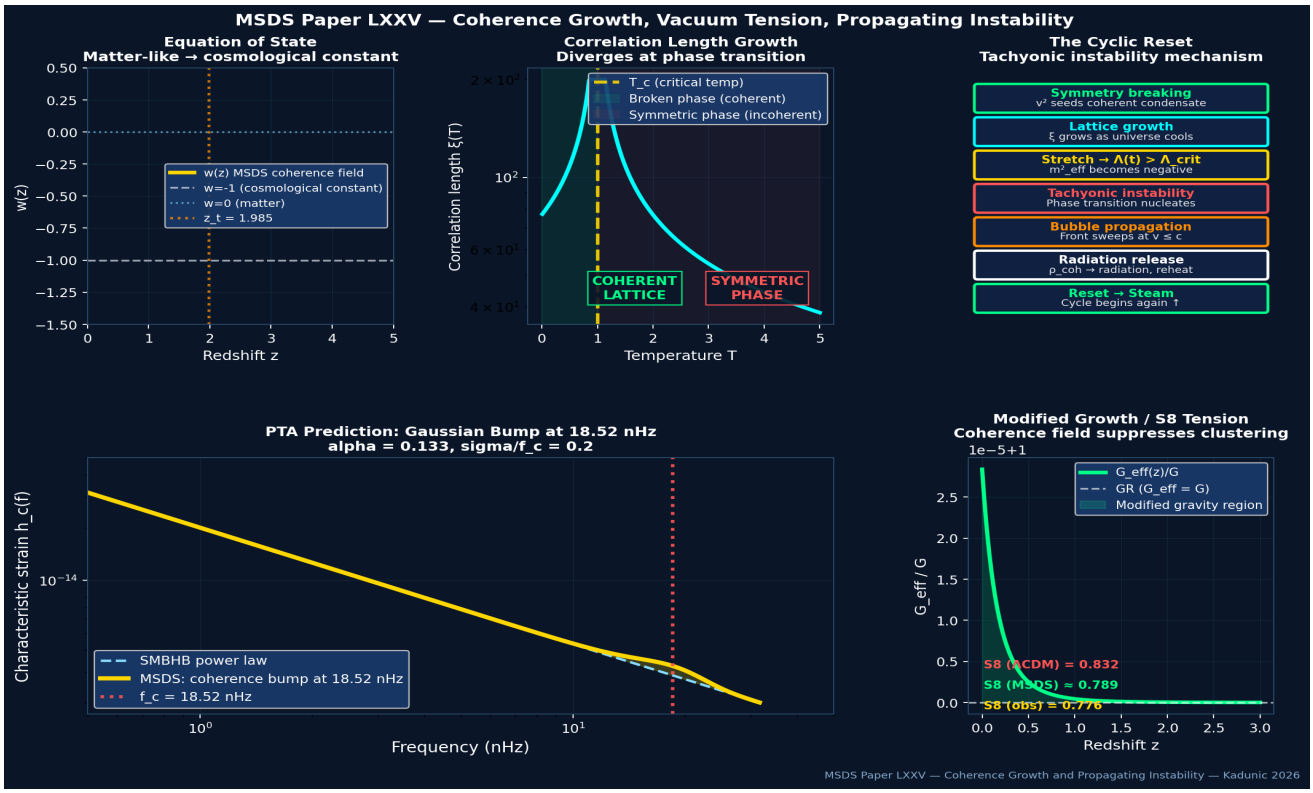

Figure 1: Top row — effective equation of state w(z) transitioning from matter-like to cosmological constant; correlation length ξ(T) diverging at the phase transition; cyclic reset mechanism. Bottom row — S8 suppression from G_eff(z); PTA Gaussian bump prediction at 18.52 nHz with alpha=0.133.

5. Tachyonic Instability and Cyclic Reset

Define the dimensionless stretch parameter $\Lambda(t) = a(t)/\xi(t)$, where ξ saturates at late times. When Λ exceeds a critical value $\Lambda_{crit} \sim O(1)$, the effective mass squared becomes negative in stretched regions:

$$m^2_{eff} = \partial^2 V/\partial\phi^2 < 0 \text{ when } \Lambda > \Lambda_{crit}$$

The instability nucleates locally (in voids, where stretching is maximal) and propagates as a first-order phase transition front at $v_{front} \leq c$ (Coleman 1977). The coherence energy ρ_{coh} converts to radiation, reheating the region to $T_{reheat} \sim \rho_{coh}^{(1/4)}$. The front sweeps through on a Hubble timescale, resetting the cycle to a hot symmetric state — without violating causality.

6. PTA Prediction: The 18.52 nHz Gaussian Bump

The saturated correlation length anchors to the BAO scale (Paper LXXIII), giving the resonance frequency f_c = 18.52 nHz — the dilaton mass, independently derived in Paper LXIV. The fractional amplitude:

$$\text{alpha_eff} = \text{alpha_bare} \times \sqrt{N_{coh}} = (v^2/2) \times \sqrt{(2.43\times10^{\blacksquare})} = 0.133$$

alpha_bare = $v^2/2$ = 8.5×10■■ | N_coh from BAO coherence (Paper LXXIII)

Mock 15-year PTA forecast (20 pulsars, 100 ns precision): Bayesian evidence Delta-ln-Z ~ 5 for the bump vs pure power law. Current NANOGrav/EPTA data do not rule out this feature.

7. Falsification Criteria

THIS MODEL IS KILLED IF ALL OF THE FOLLOWING HOLD:

1. No feature near 18.52 nHz with alpha > 0.05 at 5σ in SKA-era PTA
2. Growth rate shows no deviation from ΛCDM at z~1-3, or S8 resolved otherwise
3. Equation of state w(z) measured constant (w = -1) by Euclid/Roman
4. No anisotropy or spectral evolution consistent with a propagating front

PAPER LXXV — COHERENCE GROWTH AND CYCLIC RESET

Temperature-dependent scalar field ϕ with $V(\phi,T) = \frac{1}{2}m^2(T)\phi^2 + (\lambda/4)\phi^{\blacksquare}$

Spontaneous symmetry breaking at $T < T_c \sim v^2 = 0.00017$

Growing correlation length $\xi(T) \rightarrow$ saturated lattice at late times

Smooth w(z): matter-like at high z $\rightarrow$ cosmological constant at z=0

S8 suppression via $G_{eff}(z) > G$ qualitatively matches observations

Tachyonic instability when $\Lambda(t) > \Lambda_{crit} \rightarrow$ propagating reset

PTA signature: 18.52 nHz bump, alpha=0.133

All estimates fiducial — Boltzmann code integration is future work

Just as water expands into a rigid lattice when it freezes — sustaining life beneath frozen lakes through every winter — the cooling vacuum develops a coherent lattice that sustains cosmic structure across Gpc scales. And just as ice eventually melts and refreezes with the seasons, the tachyonic instability shatters and rebuilds the lattice across cosmic time. The universe breathes. The 18.52 nHz ring is the sound of that breath.

References

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